

CSE-307 NETWORKING LAB REPORT

On

5-Floor Network Design for CS Info-tech

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Physical Scenario Creation:

CS Info-tech is a midsize enterprise with a 5-floor office building. Each floor has a different number of computers. The aim is to design a scalable, efficient, and reliable network infrastructure. The servers are distributed strategically across floors for optimized communication and service availability. Cisco Packet Tracer is used for simulation.

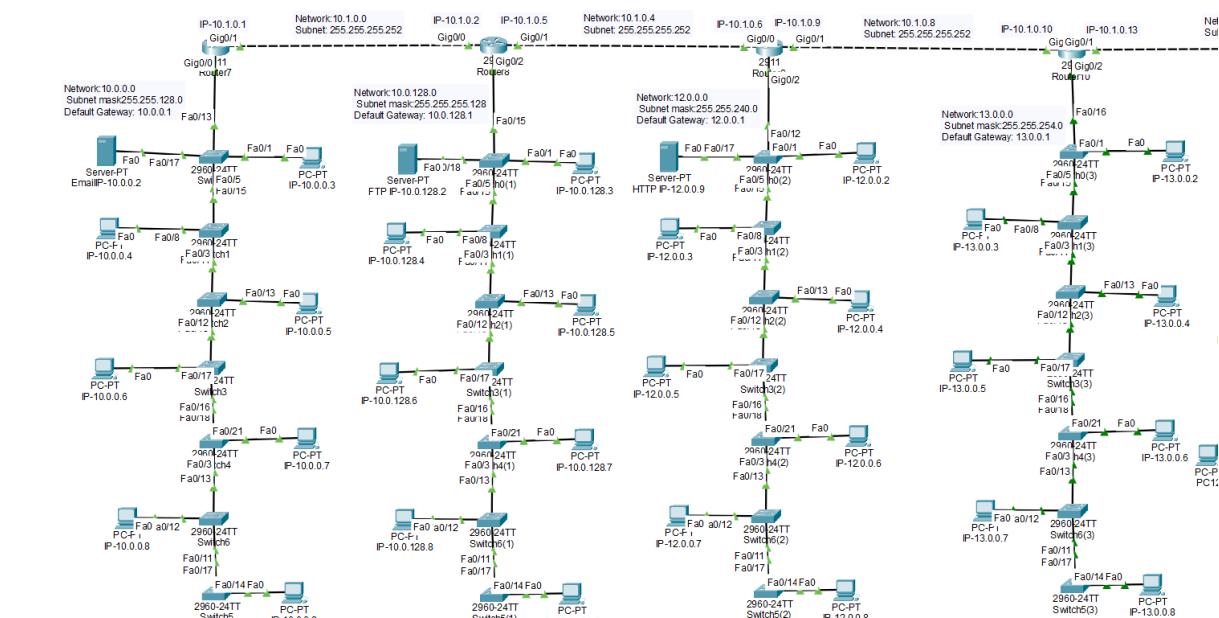


Image-1. Floors 1-4

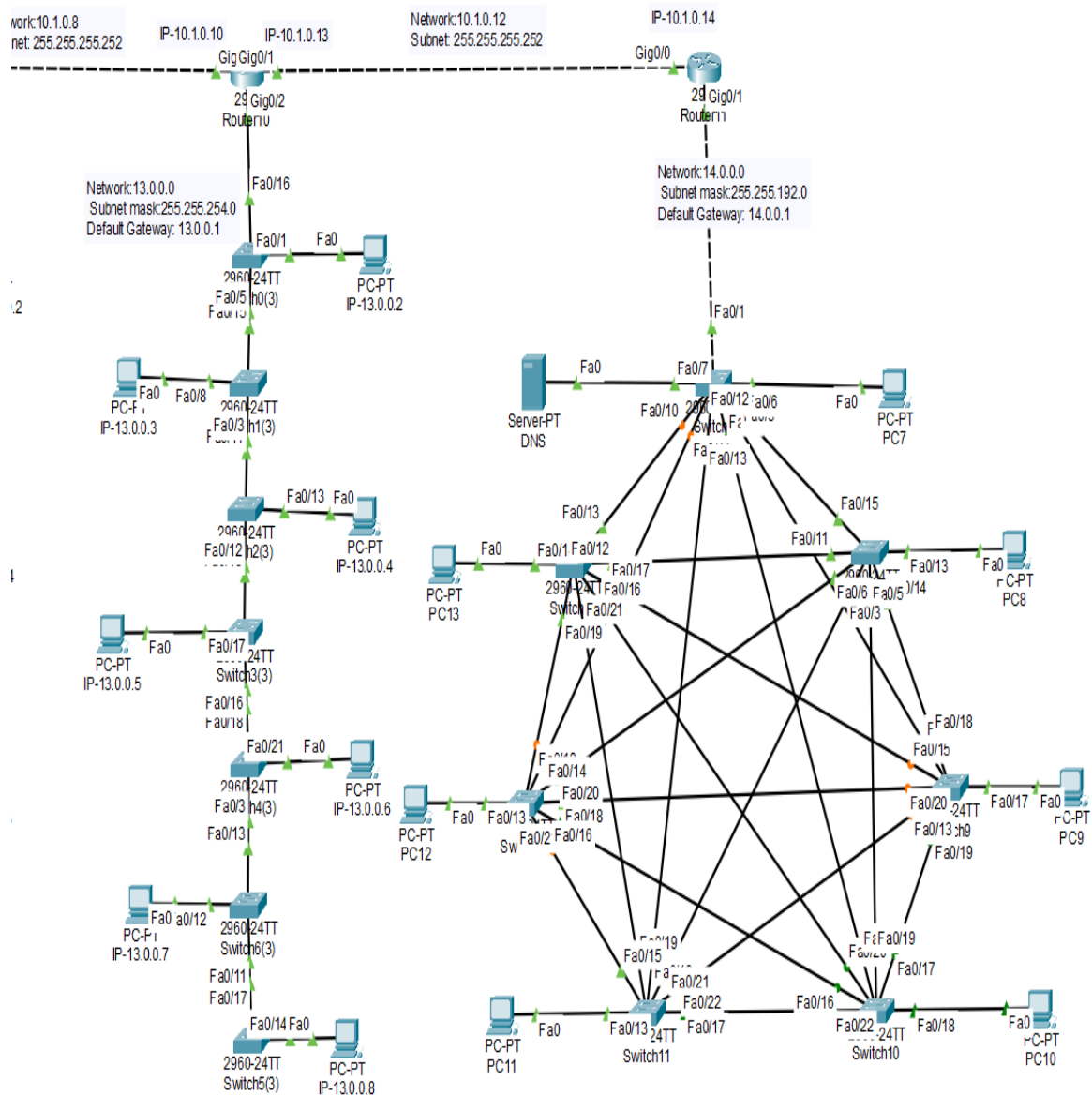


Image-2. Floors 5

- Switch used - 2960
- Router used -2911
- Floor 1 to Floor 4 – Bus topology
- Floor 5 – Mesh Topology

IP Addressing Scheme

Following the classless addressing scheme:

Floor	Class	Network Address	Gateway Address
Floor 1	Class A Private	10.0.0.0/17	255.255.128.0
Floor 2	Class A Private	10.0.128.0/25	255.255.255.128
Floor 3	Class A Public	12.0.0.0/20	255.255.240.0
Floor 4	Class A Public	13.0.0.0/23	255.255.254.0
Floor 5	Class A Public	14.0.0.0/18	255.255.192.0

Table 1. Floor-wise Gateway and IP configuration

Floor 1 IP Configuration ☐ DHCP ☒ Static IPv4 Address 10.0.0.3 Subnet Mask 255.255.128.0 Default Gateway 10.0.0.1 DNS Server 0.0.0.0	Floor 2 IP Configuration ☐ DHCP ☒ Static IPv4 Address 10.0.128.3 Subnet Mask 255.255.255.128 Default Gateway 10.0.128.1 DNS Server 0.0.0.0
Floor 3 IP Configuration ☐ DHCP ☒ Static IPv4 Address 12.0.0.2 Subnet Mask 255.255.240.0 Default Gateway 12.0.0.1 DNS Server 0.0.0.0	Floor 4 IP Configuration ☐ DHCP ☒ Static IPv4 Address 13.0.0.2 Subnet Mask 255.255.254.0 Default Gateway 13.0.0.1 DNS Server 0.0.0.0
Floor 5 IP Configuration ☒ DHCP ☐ Static IPv4 Address 14.0.0.9 Subnet Mask 255.255.192.0 Default Gateway 14.0.0.1 DNS Server 14.0.0.2	

Table 2. Snaps of IP configuration window from pc of each floor

Router IP Configuration

```
-----_--_-----
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 7 subnets, 3 masks
R    10.0.0.0/17 [120/1] via 10.1.0.1, 00:00:12, GigabitEthernet0/0
C    10.1.0.0/30 is directly connected, GigabitEthernet0/0
L    10.1.0.2/32 is directly connected, GigabitEthernet0/0
C    10.2.0.4/30 is directly connected, GigabitEthernet0/1
L    10.2.0.5/32 is directly connected, GigabitEthernet0/1
R    10.3.0.8/30 [120/1] via 10.2.0.6, 00:00:21, GigabitEthernet0/1
R    10.4.0.12/30 [120/2] via 10.2.0.6, 00:00:21, GigabitEthernet0/1
R    12.0.0.0/8 [120/1] via 10.2.0.6, 00:00:21, GigabitEthernet0/1
R    13.0.0.0/8 [120/2] via 10.2.0.6, 00:00:21, GigabitEthernet0/1
R    14.0.0.0/8 [120/3] via 10.2.0.6, 00:00:21, GigabitEthernet0/1

Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/2
Router(config-if)#ip address 10.0.128.1 255.255.255.252
Router(config-if)#ip address 10.0.128.1 255.255.255.252
Router(config-if)#ip address 10.0.128.1 255.255.255.128
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/2
Router(config-if)#
```

Image 3. Ex.IP addressing of Floor 5's router in CLI

Routing

In this project, we used RIP (Routing Information Protocol) for dynamic routing between the floor routers. RIP was configured on each floor's router, allowing the routers to automatically exchange routes and learn about networks on other floors.

This makes sure that devices from one floor can communicate with devices on any other floor without needing manual static routes for each connection.

Floor 1 <pre> Router(config-if)# Router(config-if)#router rip Router(config-router)#network 10.0.0.0 Router(config-router)#network 10.1.0.0 Router(config-router)#version 2 Router(config-router)#exit Router(config)#exit Router# </pre>
Floor 2 <pre> Router#en Router#conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#router rip Router(config-router)#network 10.1.0.0 Router(config-router)#network 10.0.128.0 Router(config-router)#network 10.2.0.4 Router(config-router)#eit ^ % Invalid input detected at '^' marker. Router(config-router)#exit Router(config)#exit _ </pre>
Floor 3 <pre> Router# Router#conf t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#router rip Router(config-router)#network 10.0.128.0 Router(config-router)#network 10.1.0.0 Router(config-router)#network 10.2.0.0 Router(config-router)#exit Router(config)#exit </pre>
Floor 4 <pre> Router# Router#config t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#router rip Router(config-router)#network 10.3.0.8 Router(config-router)#network 10.4.0.12 Router(config-router)#network 13.0.0.0 Router(config-router)#version 2 Router(config-router)# </pre>
Floor 5 <pre> Router# Router#config t Enter configuration commands, one per line. End with CNTL/Z. Router(config)#router rip Router(config-router)#network 10.4.0.12 Router(config-router)#network 14.0.0.0 Router(config-router)#version 2 Router(config-router)#exit Router(config)# </pre>

Table 3. Snaps of dynamic routing between Routers using rip command on CLI

Communication between PC's

To check communication between pc's we used the ping command in the command prompt and to see the paths we used tracert command also to see flow of packets.

Floor 1 to 2

```
C:\>ping 10.0.128.3

Pinging 10.0.128.3 with 32 bytes of data:

Reply from 10.0.128.3: bytes=32 time<1ms TTL=126
Reply from 10.0.128.3: bytes=32 time=1ms TTL=126
Reply from 10.0.128.3: bytes=32 time<1ms TTL=126
Reply from 10.0.128.3: bytes=32 time<1ms TTL=126

Ping statistics for 10.0.128.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Floor 1 to 3

```
C:\>ping 12.0.0.2

Pinging 12.0.0.2 with 32 bytes of data:

Request timed out.
Reply from 12.0.0.2: bytes=32 time=1ms TTL=125
Reply from 12.0.0.2: bytes=32 time<1ms TTL=125
Reply from 12.0.0.2: bytes=32 time<1ms TTL=125

Ping statistics for 12.0.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Floor 1 to 4

```
C:\>ping 13.0.0.2

Pinging 13.0.0.2 with 32 bytes of data:

Reply from 13.0.0.2: bytes=32 time<1ms TTL=124
Reply from 13.0.0.2: bytes=32 time=20ms TTL=124
Reply from 13.0.0.2: bytes=32 time<1ms TTL=124
Reply from 13.0.0.2: bytes=32 time<1ms TTL=124

Ping statistics for 13.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 20ms, Average = 5ms

C:\>
```

Floor 1 to 5

```
C:\>ping 14.0.0.9

Pinging 14.0.0.9 with 32 bytes of data:

Reply from 14.0.0.9: bytes=32 time<1ms TTL=123
Reply from 14.0.0.9: bytes=32 time<1ms TTL=123
Reply from 14.0.0.9: bytes=32 time<1ms TTL=123
Reply from 14.0.0.9: bytes=32 time<1ms TTL=123

Ping statistics for 14.0.0.9:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Table 4. Snaps of inter-communication between PC's of all floors Using the ping command


```

C:\>tracert 14.0.0.9

Tracing route to 14.0.0.9 over a maximum of 30 hops:

  1    0 ms      22 ms      0 ms      10.0.0.1
  2    0 ms      0 ms       0 ms      10.1.0.2
  3    0 ms      0 ms      28 ms     10.2.0.6
  4    0 ms      0 ms       0 ms     10.3.0.10
  5    0 ms      0 ms       0 ms     10.4.0.14
  6    0 ms      0 ms       0 ms     14.0.0.9

Trace complete.

C:\>

```

Image 4. Use of Tracert command on Floor 1's PC to Floor 7's PC to see flow of data to trace the route.

Server Placement

Server Type	Location	IP Address Example
Email	Floor 1	10.0.0.2
FTP	Floor 2	10.0.128.2
HTTP	Floor 3	12.0.0.9
DNS & DHCP	Floor 5	14.0.0.2

Table 5 shows the Servers and Their IP Addresses.


```

C:\>ftp
C:\>ftp 10.0.128.2
Trying to connect...10.0.128.2
Connected to 10.0.128.2
220- Welcome to PT Ftp server
Username:nissi
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>dir

Listing /ftp directory from 10.0.128.2:
 0  : c1.bin                               5571584
 1  : c1841-advipservicesk9-mz.124-15.T1.bin 33591768
 2  : c1841-ipbase-mz.123-14.T7.bin         13832032
 3  : c1841-ipbasek9-mz.124-12.bin          16599160
 4  : c1900-universalk9-mz.SPA.155-3.M4a.bin 33591768
 5  : c2600-advipservicesk9-mz.124-15.T1.bin 33591768
 6  : c2600-ipbasek9-mz.124-8.bin           13169700
 7  : c2800nm-advipservicesk9-mz.124-15.T1.bin 50938004
 8  : c2800nm-advipservicesk9-mz.151-4.M4.bin 33591768
 9  : c2800nm-ipbase-mz.123-14.T7.bin       5571584
10  : c2800nm-ipbasek9-mz.124-8.bin         15522644
11  : c2900-universalk9-mz.SPA.155-3.M4a.bin 33591768
12  : c2950-i6q4l2-mz.121-22.EA4.bin       3058048
13  : c2950-i6q4l2-mz.121-22.EA8.bin       3117390
14  : c2960-lanbase-mz.122-25.FX.bin        4414921
15  : c2960-lanbase-mz.122-25.SEE1.bin     4670455
16  : c2960-lanbasek9-mz.150-2.SE4.bin     4670455
17  : c3560-advipservicesk9-mz.122-37.SE1.bin 8662192
18  : c3560-advipservicesk9-mz.122-46.SE1.bin 10713279
19  : c800-universalk9-mz.SPA.152-4.M4.bin  33591768
20  : c800-universalk9-mz.SPA.154-3.M6a.bin 83029236
21  : cat3k_caa-universalk9.16.03.02.SPA.bin 505532849
22  : cgr1000-universalk9-mz.SPA.154-2.CG   159487552
23  : cgr1000-universalk9-mz.SPA.156-3.CG   184530138
24  : ir800-universalk9-bundle.SPA.156-3.M.bin 160968869
25  : ir800-universalk9-mz.SPA.155-3.M     61750062
26  : ir800-universalk9-mz.SPA.156-3.M     63753767
27  : ir800_yocto-1.7.2.tar                2877440
28  : ir800_yocto-1.7.2.python-2.7.3.tar   6912000
29  : pt1000-i-mz.122-28.bin               5571584
30  : pt3000-i6q4l2-mz.121-22.EA4.bin      3117390

```

```

ftp>rename c1.bin nissi.bin

Renaming c1.bin
ftp>
[OK Renamed file successfully from c1.bin to nissi.bin]
ftp>get nissi.bin

Reading file nissi.bin from 10.0.128.2:
File transfer in progress...

[Transfer complete - 5571584 bytes]

5571584 bytes copied in 27.002 secs (47278 bytes/sec)
ftp>delete nissi.bin

Deleting file nissi.bin from 10.0.128.2: ftp>
[Deleted file nissi.bin successfully ]
ftp>

```

```

C:\>dir

Volume in drive C has no label.
Volume Serial Number is 5E12-4AF3
Directory of C:\

1/1/1970    5:30 PM                5571584    nissi.bin
1/1/1970    5:30 PM                  26         sampleFile.txt
               5571610 bytes          2 File(s)

C:\>ftp
C:\>ftp 10.0.128.2
Trying to connect...10.0.128.2
Connected to 10.0.128.2
220- Welcome to PT Ftp server
Username:nissi
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>put sampleFile.txt

Writing file sampleFile.txt to 10.0.128.2:
File transfer in progress...

[Transfer complete - 26 bytes]

26 bytes copied in 0.048 secs (541 bytes/sec)
ftp>

```

Table 7- shows the working of FTP Server in the Second Floor

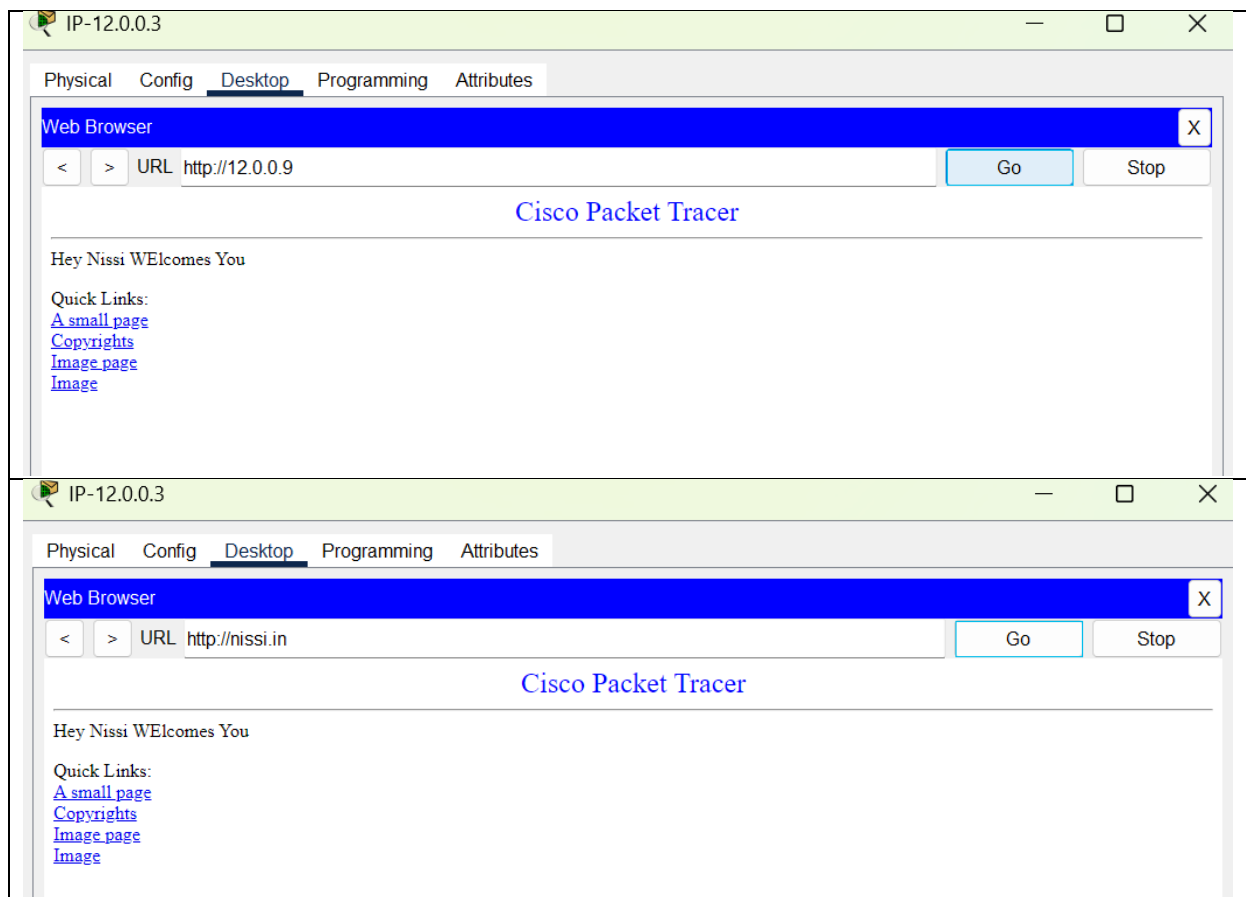


Table-8 shows the Working of the HTTP Server

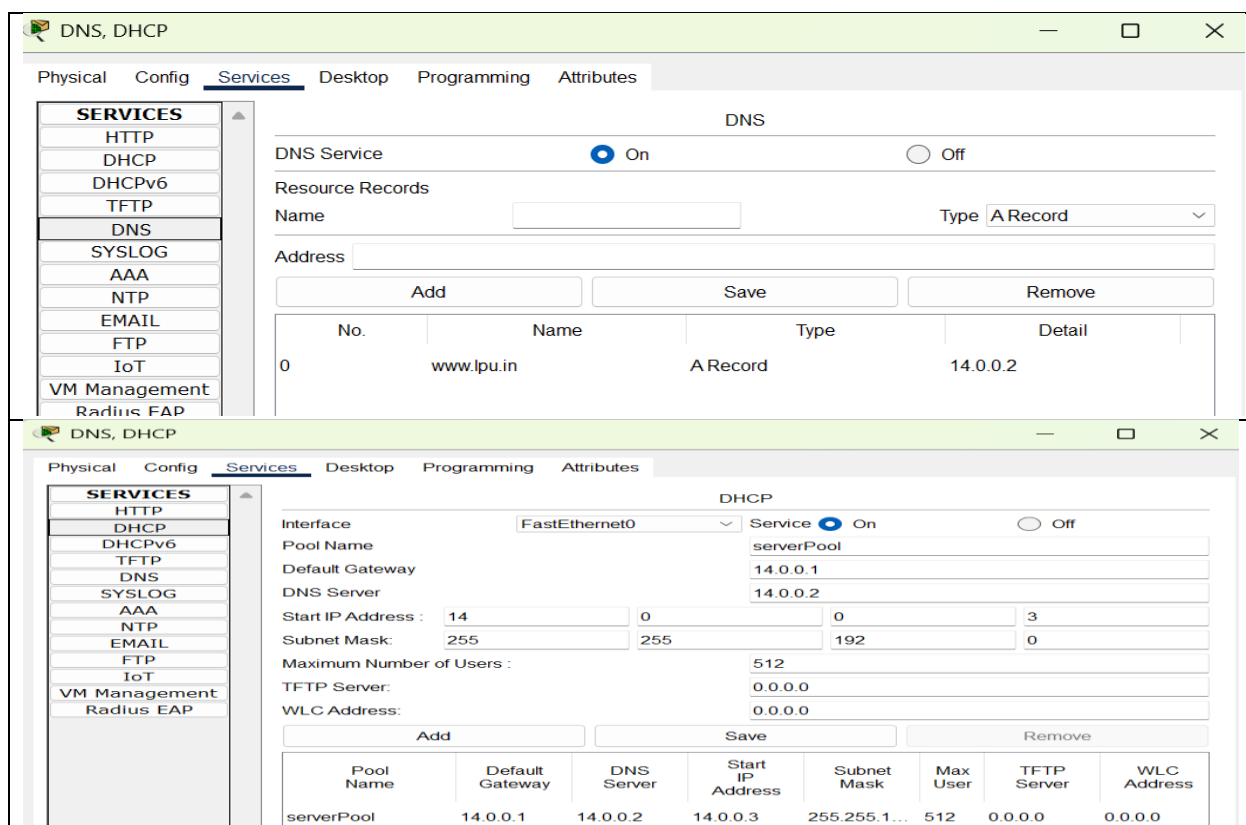


Table-9 shows the Working of Both the DNS and DHCP Server

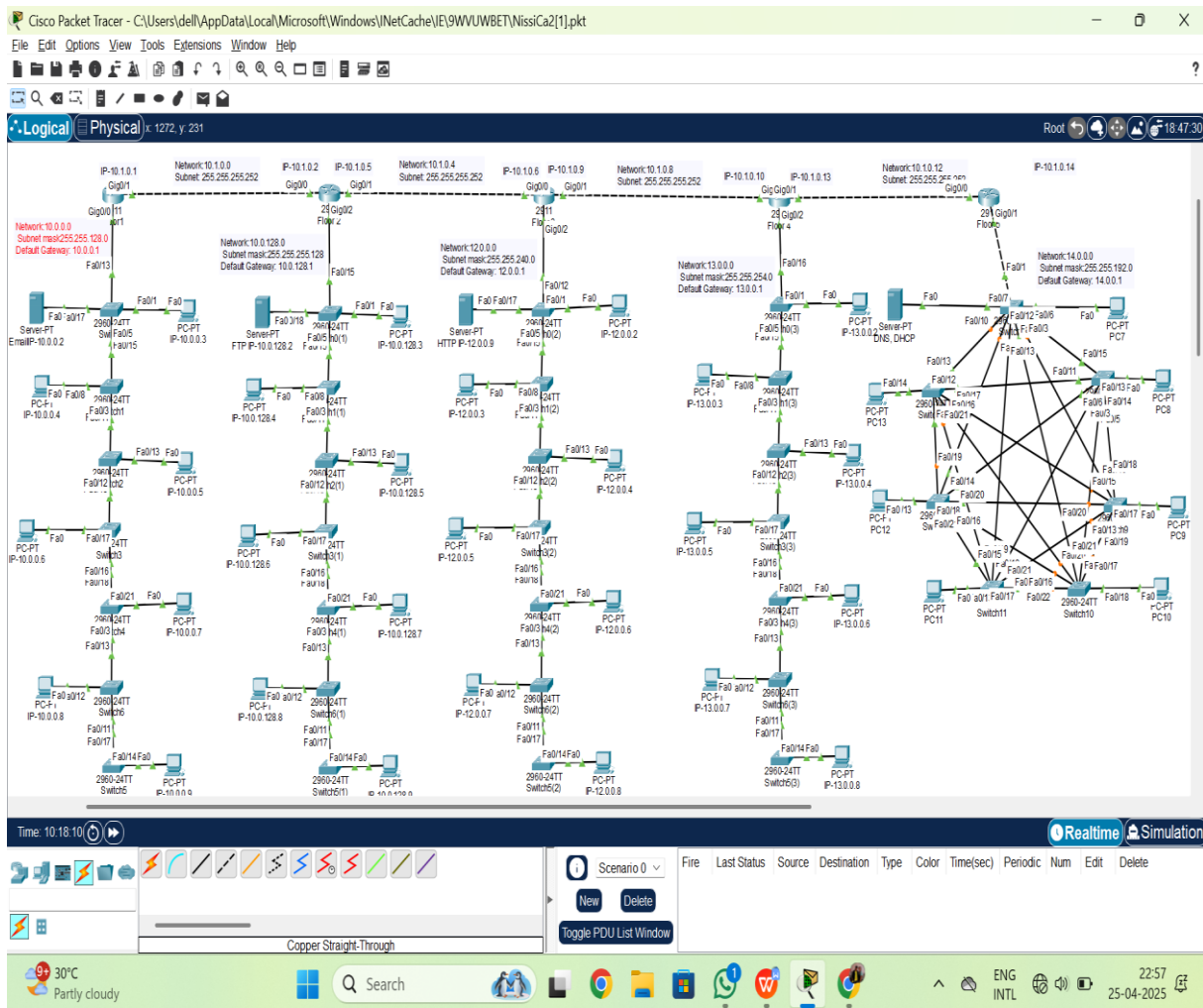


Image-5 Shows is the whole Representation of the Network with all the Port Labels in Work Space

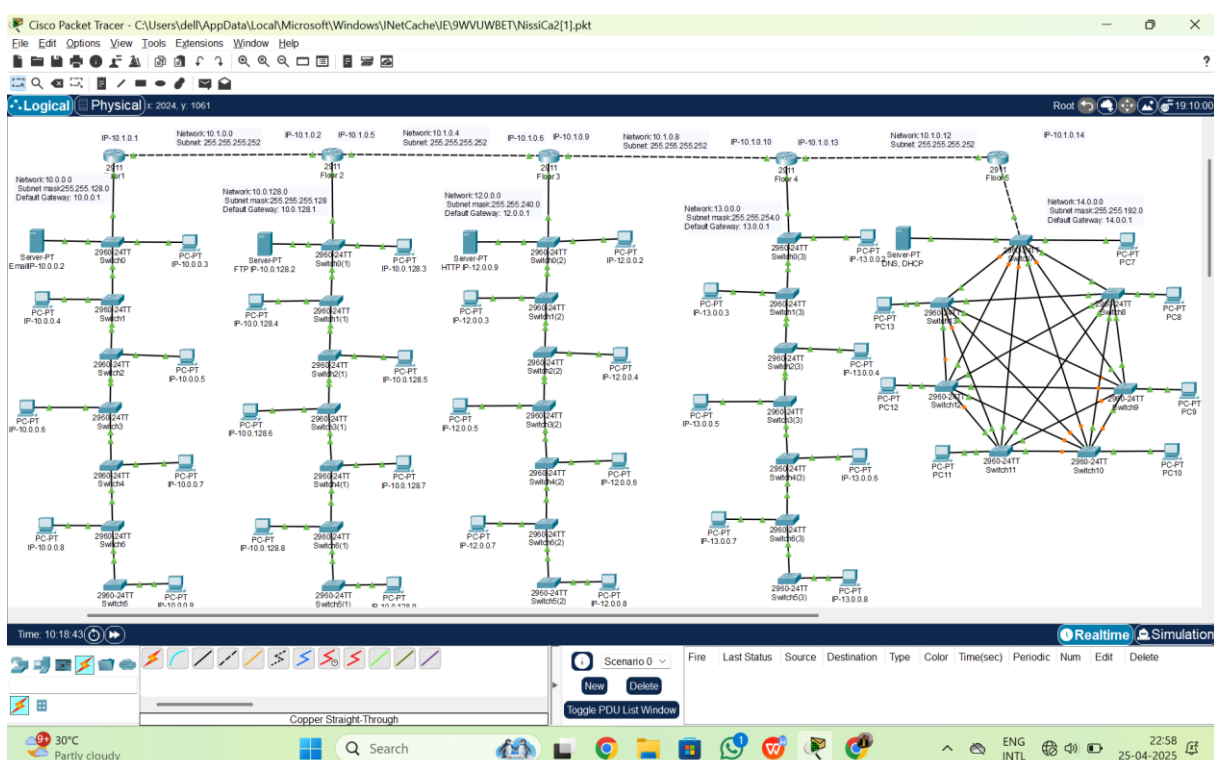


Image-6 Shows is the whole Representation of the Network